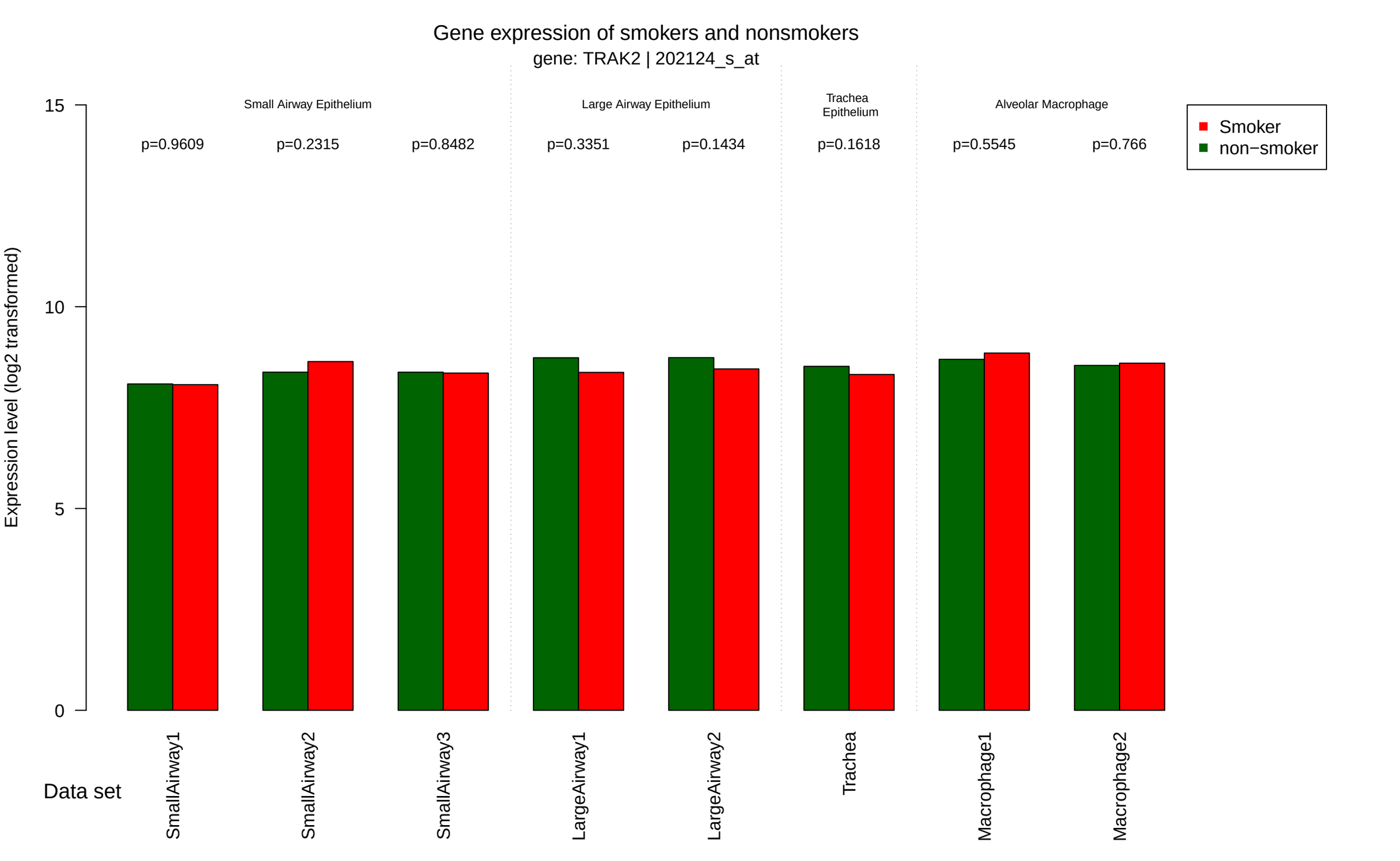
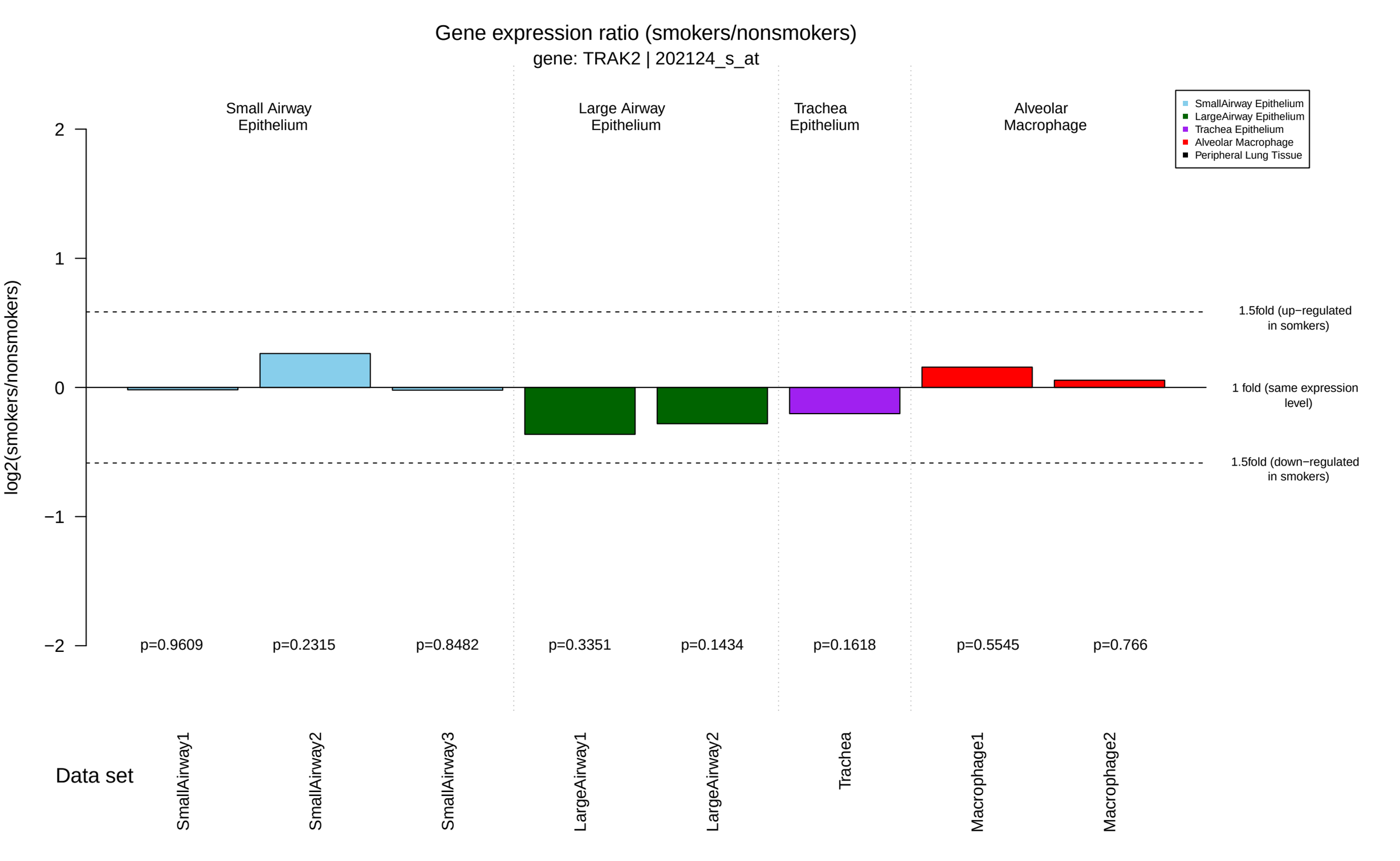
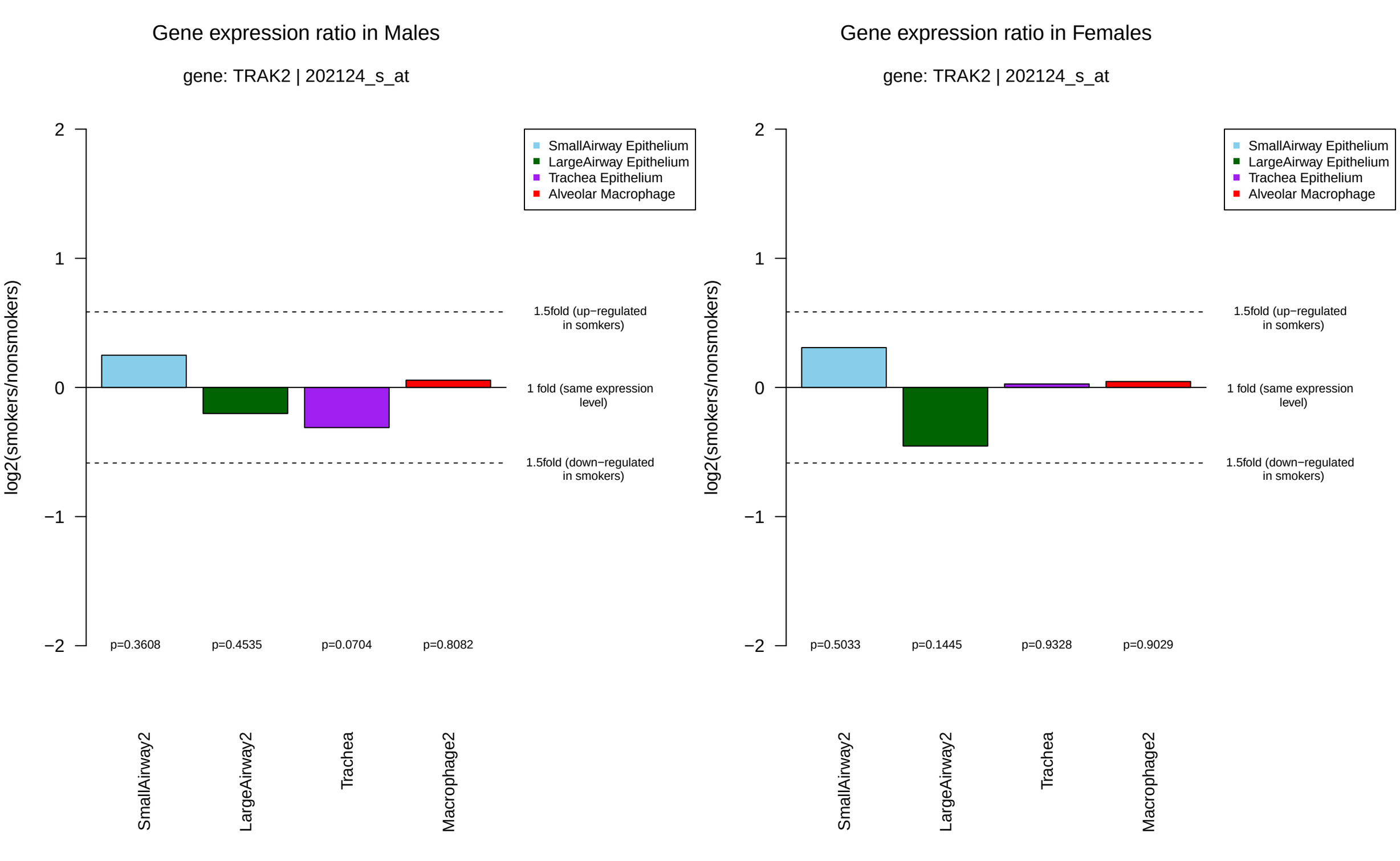
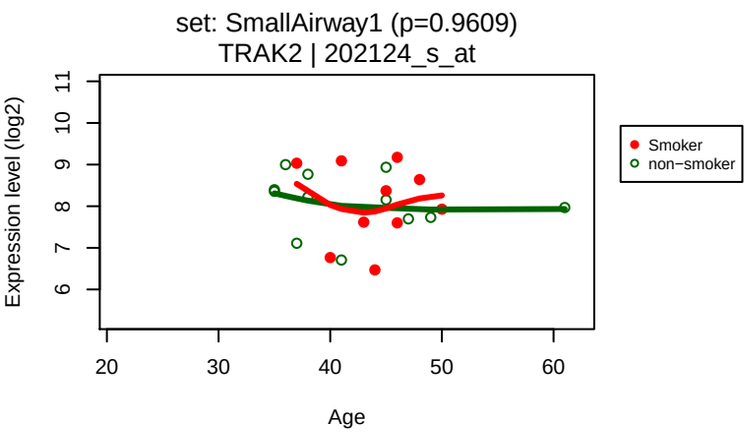
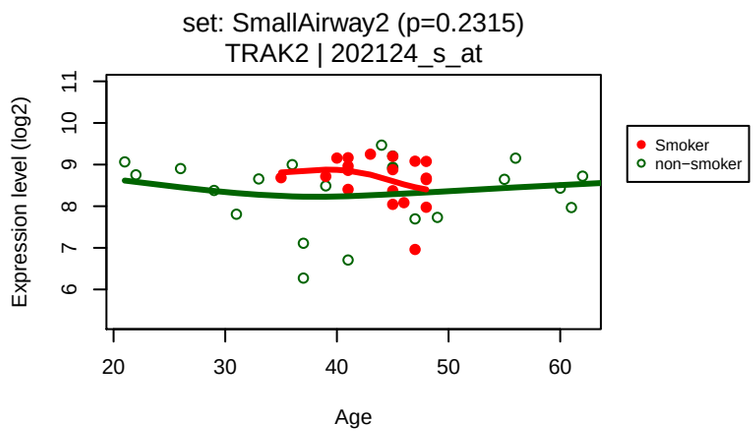
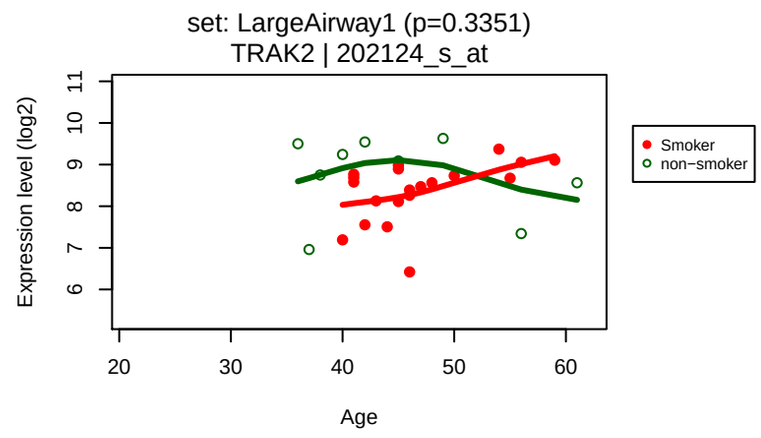

Gene expression of smokers and nonsmokers
gene: TRAK2 | 202124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9609
p=0.2315
p=0.8482
p=0.3351
p=0.1434
p=0.1618
p=0.5545
p=0.766
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

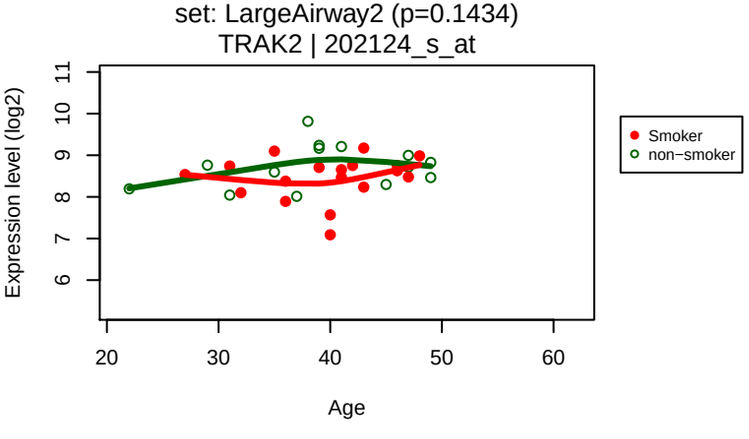

Gene expression ratio (smokers/nonsmokers)
gene: TRAK2 | 202124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9609
p=0.2315
p=0.8482
p=0.3351
p=0.1434
p=0.1618
p=0.5545
p=0.766
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

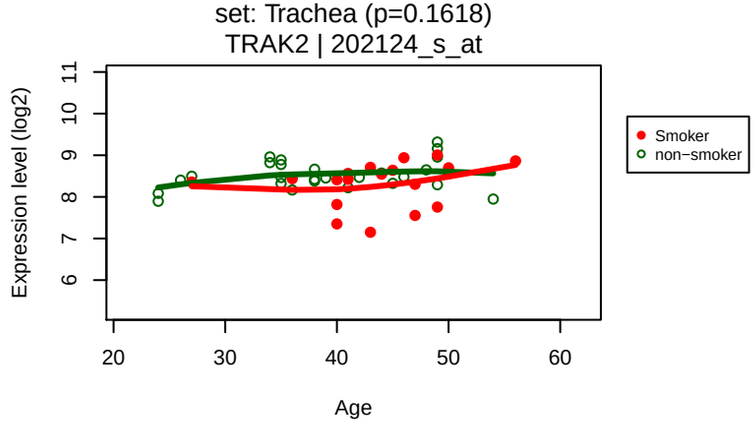

Gene expression ratio in Males
gene: TRAK2 | 202124_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3608
p=0.4535
p=0.0704
p=0.8082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK2 | 202124_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5033
p=0.1445
p=0.9328
p=0.9029
SmallAirway2
LargeAirway2
Trachea
Macrophage2

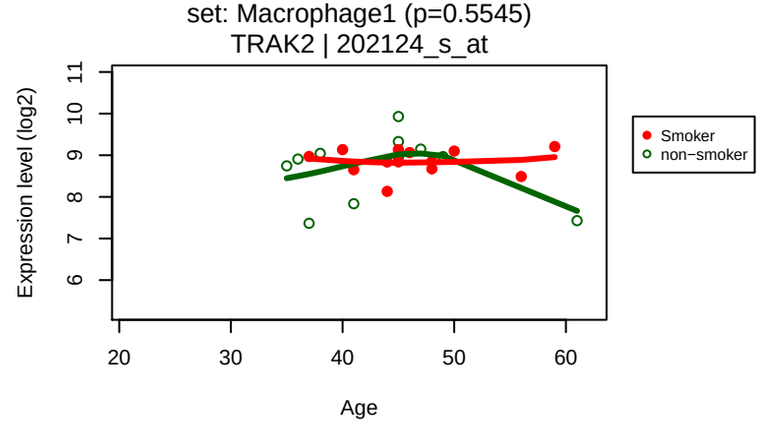

set: SmallAirway1 (p=0.9609)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker

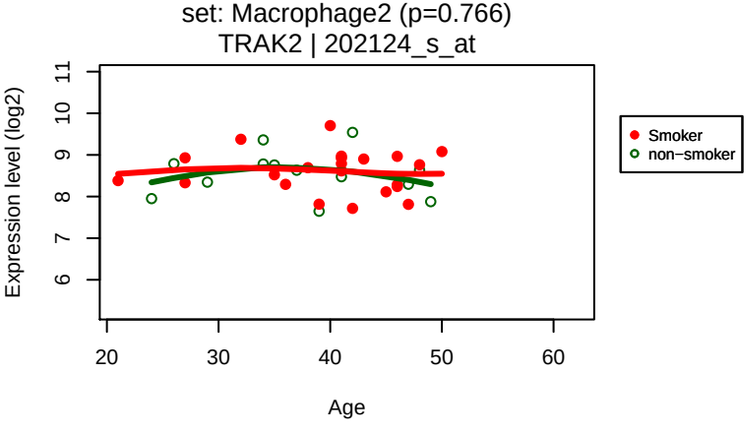

set: SmallAirway2 (p=0.2315)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker

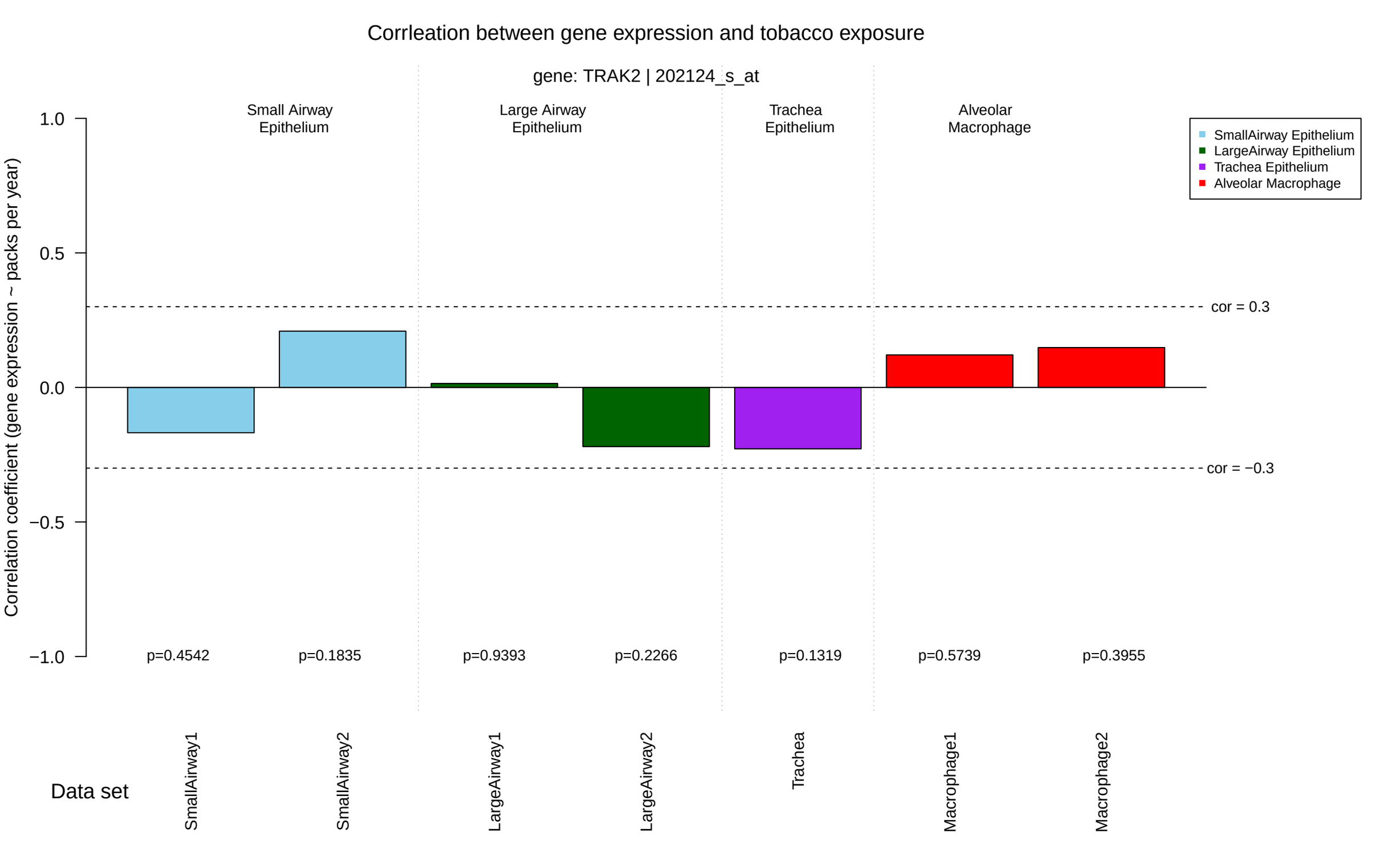

set: LargeAirway1 (p=0.3351)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1434)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1618)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5545)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.766)
TRAK2 | 202124_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK2 | 202124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4542
p=0.1835
p=0.9393
p=0.2266
p=0.1319
p=0.5739
p=0.3955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2